

Hard Optimization Problems in Mechanistic Biology

BENQODHI Workshop — BENchmarking Quantum
Optimization: from Datasets to Health Innovation

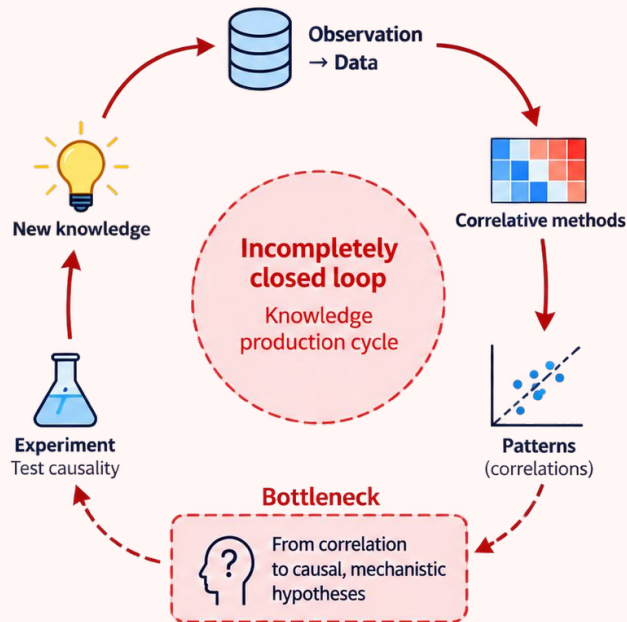
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September 10, 2026

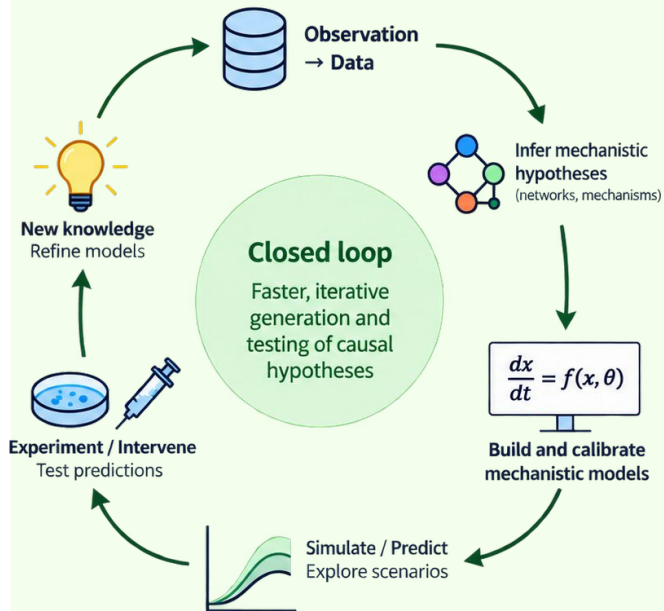
From correlation to causality

Mechanistic models turn hypotheses into, executable, testable predictions

Current situation We are very good at finding correlations, but there is a bottleneck in generating and testing causal, mechanistic hypotheses.

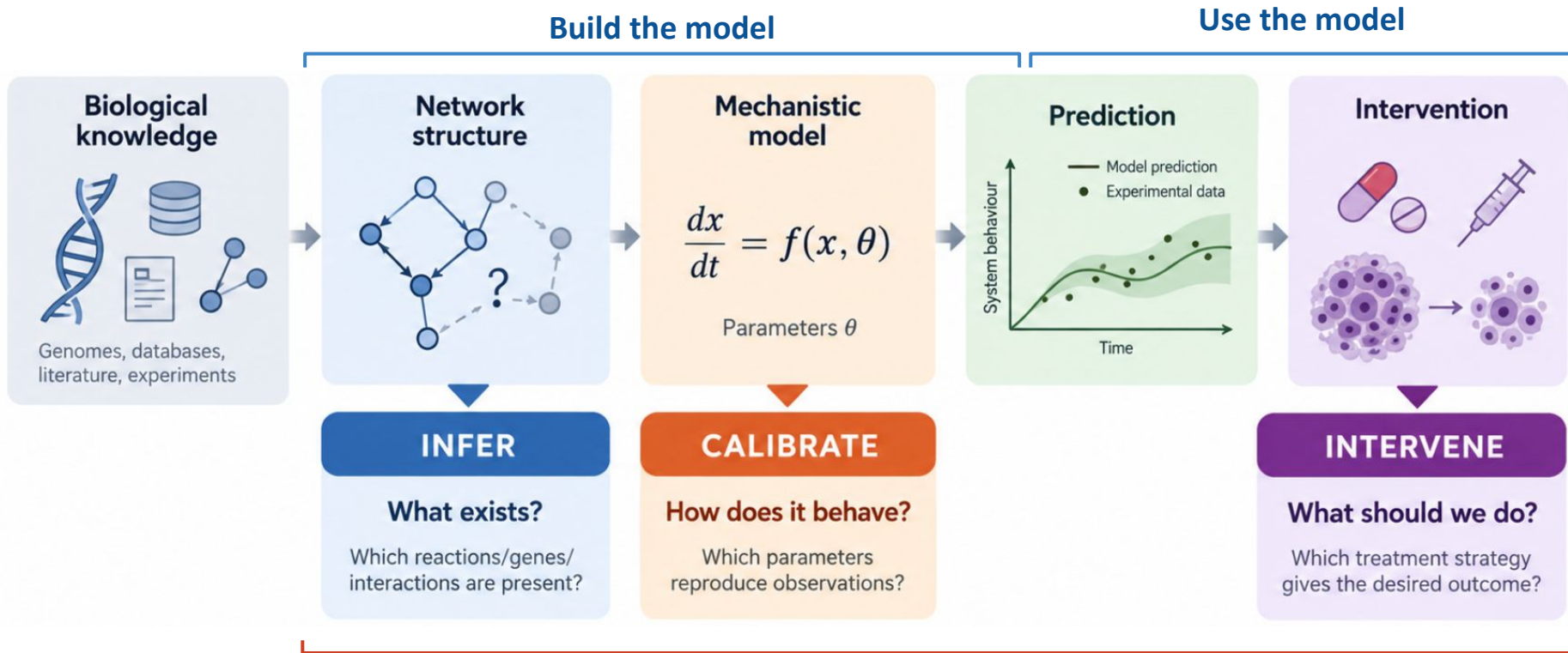


Closing the loop with mechanistic models Use mechanistic models to generate in-silico hypotheses, explore interventions and iteratively integrate new data



Mechanistic models provides a bridge from data to intervention – and every step introduce an optimization problem

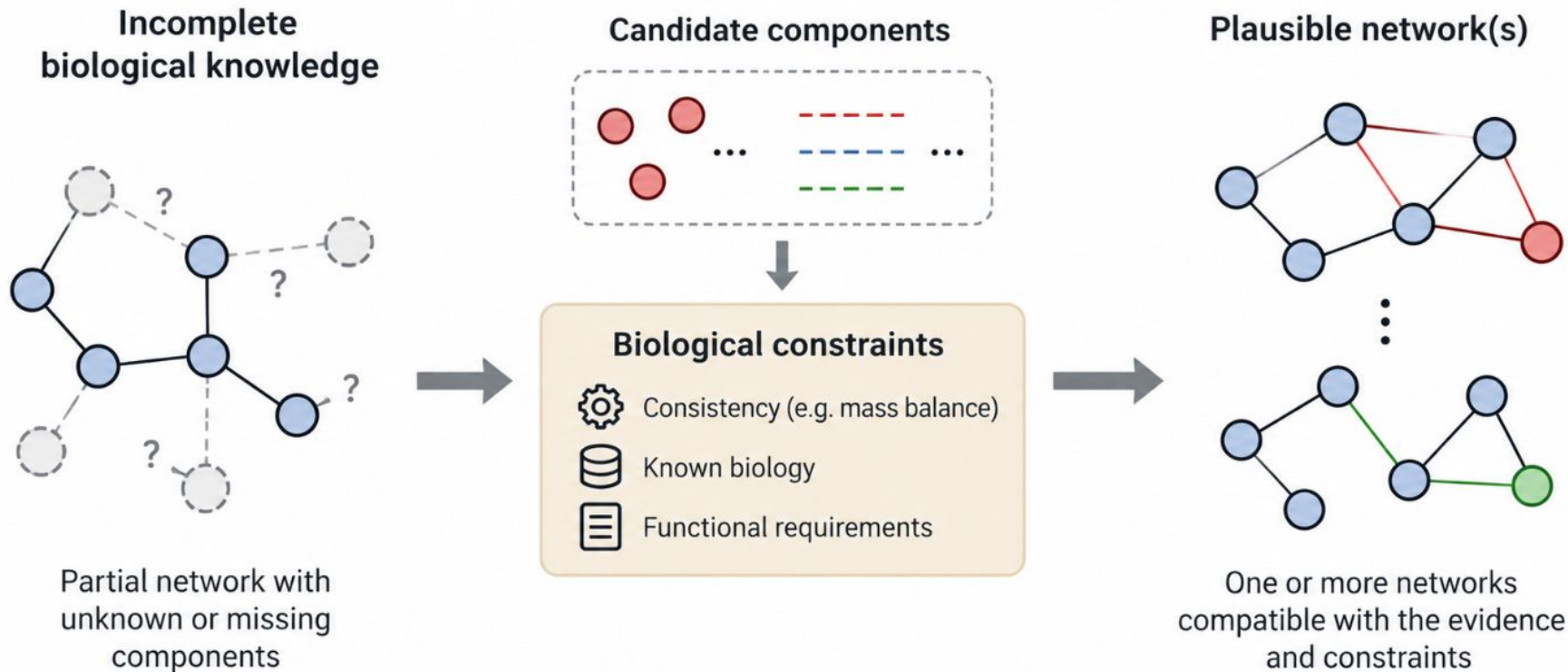
Where do hard optimization problems arise in mechanistic biology?



Different optimization problems at each step

Molecular Network Inference

What network structure is compatible with the evidence?



Constraint discrete/mixed optimization

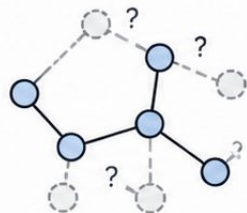
Network inference in metabolic modelling

Different biological questions, similar network-selection problem

1. Metabolic network reconstruction

From incomplete knowledge to a consistent model

Incomplete biochemical knowledge

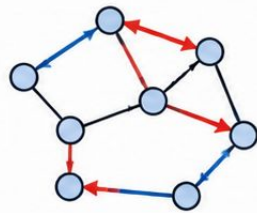


- Missing reactions
- Gaps in connectivity
- Incomplete annotation
- Orphan reactions

Candidate reactions



Consistent genome-scale model (GEM)



- Mass balanced
- Connected metabolism
- Functional capabilities
- Genotype-phenotype link

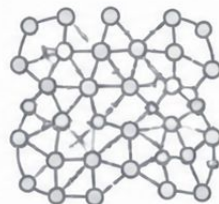
Question: Which reactions should be added?

What could be there?

2. Context-specific model reconstruction

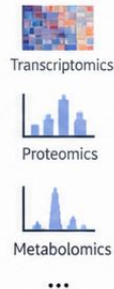
From a reference network and omics data to a context-specific model

Reference metabolic network



Genome-scale model
(e.g. AGORA, ReconX, ...)

Omics data (context)

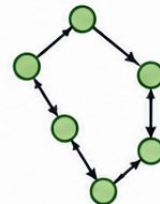


Context-specific evidence



- likely active
- likely inactive
- uncertain

Context-specific model



Question: Which part of the network is compatible with this context?

What is relevant here?

Common computational framework:

Network selection under biological constraints

MILP-based optimization | LP-based alternatives (e.g. FASTCORE)

From biological rules to genome-scale MILP

Different biological questions, similar network-selection problems

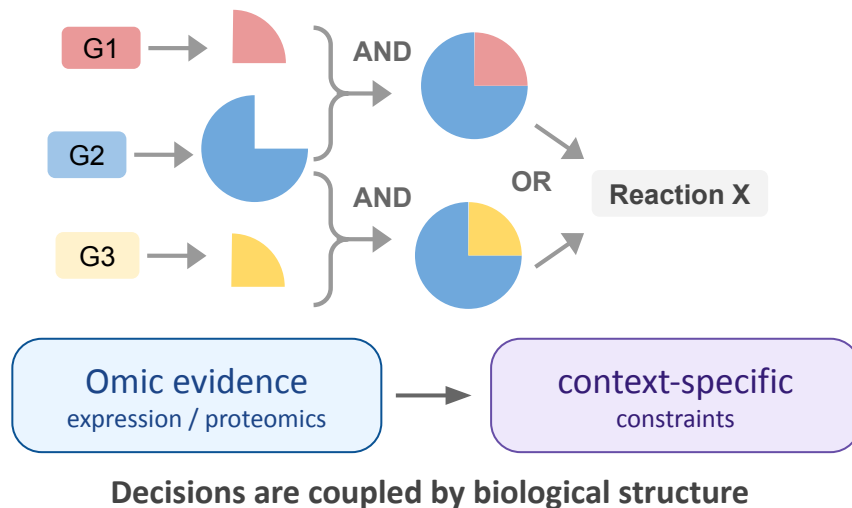
Network selection as a MILP

$$\begin{aligned}
 \min_{\mathbf{v}, \mathbf{y}} \quad & \sum_{i \in C} c_i y_i \\
 \text{s.t.} \quad & \mathbf{S} \mathbf{v} = 0 && \text{mass balance} \\
 & v_i^L y_i \leq v_i \leq v_i^U y_i && i \in C \\
 & v_i^L \leq v_i \leq v_i^U && i \notin C \\
 & \mathbf{G}(\mathbf{y}, \mathbf{x}) && \text{GPR / omic constraints} \\
 & v_{t_k} \geq v_{t_k}^{\min} && k \in \mathcal{T} \\
 & y_i \in \{0, 1\} && i \in C
 \end{aligned}$$

v_i — reaction flux y_i — binary reaction-selection variable $\mathbf{S} \cdot \mathbf{v} = 0$ — mass balance constraint v_i^L, v_i^U — reaction bounds / directionality GPR — gene-protein-reaction logic + evidence $\mathbf{x}$: evidence, e.g. gene expression v_{t_k} — required metabolic function c_i — cost for selecting reaction i

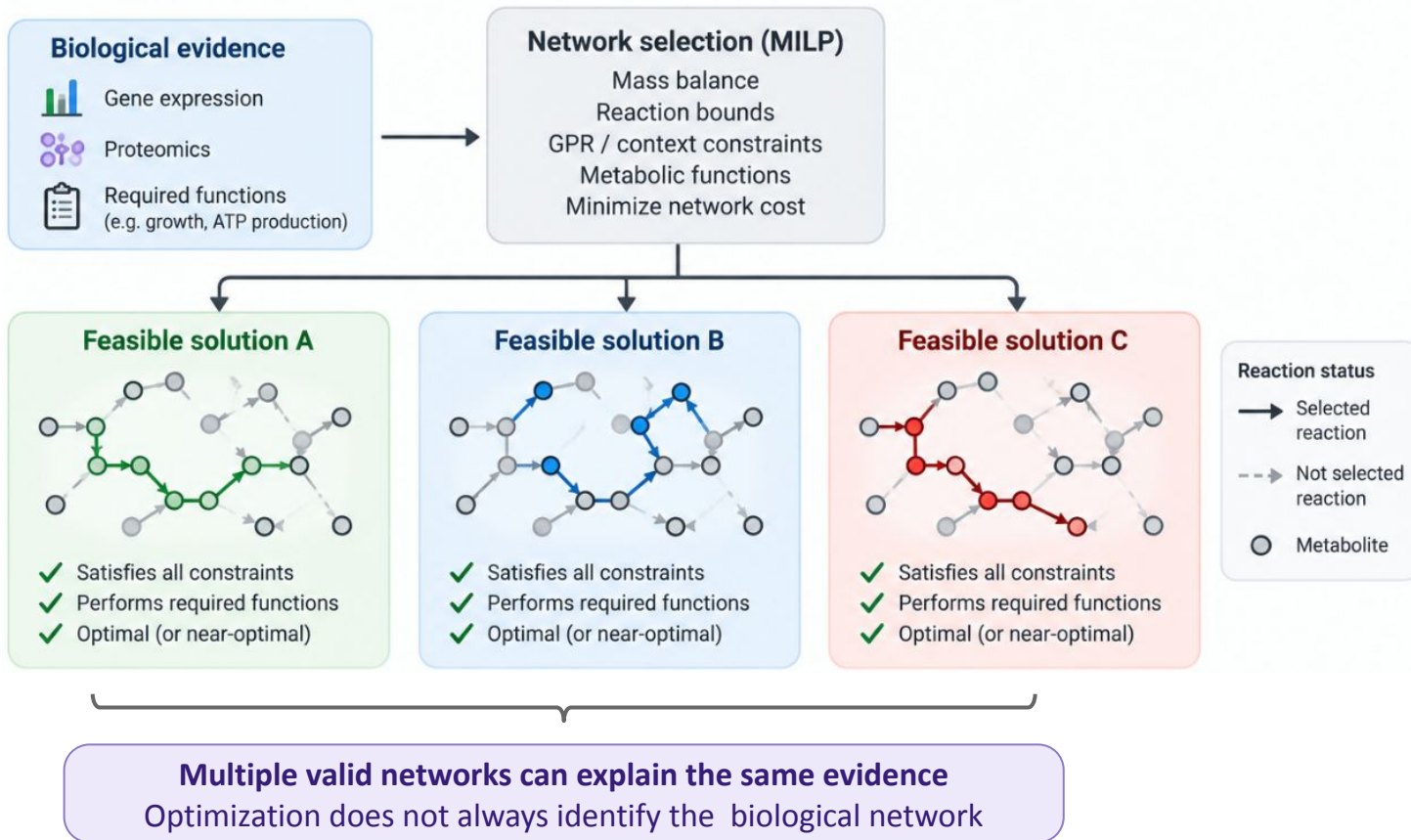
Human2: genome scale biological structure [1]

12,877 biochemical reactions 8,460 metabolites · 2,848 genes **26,246 gene-reaction pairs** Genes → GPR rules → reactions



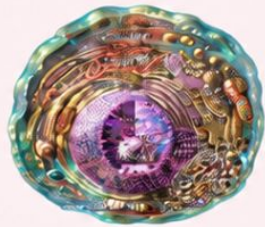
The solution is not necessarily unique

Different networks can satisfy the same biological constraints



From networks to multiscale systems

Biological dynamics span multiple spatial and temporal scales



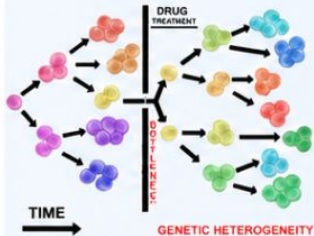
Cell complexity

- Metabolism and regulation
- Multiple cellular processes
- Genetic alterations
- Phenotypic diversity



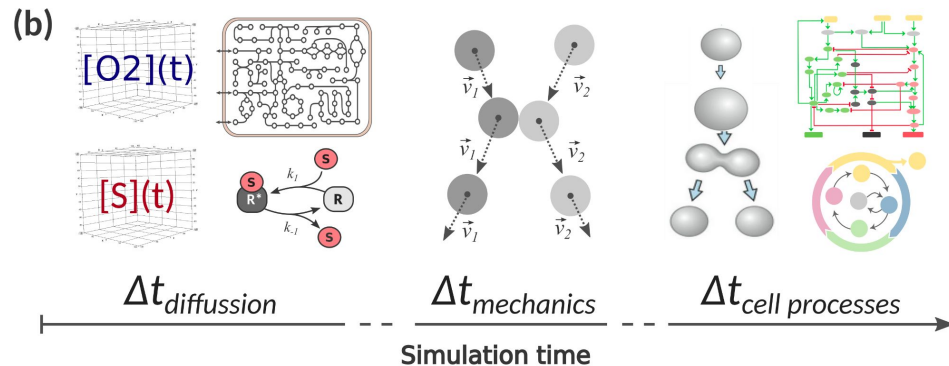
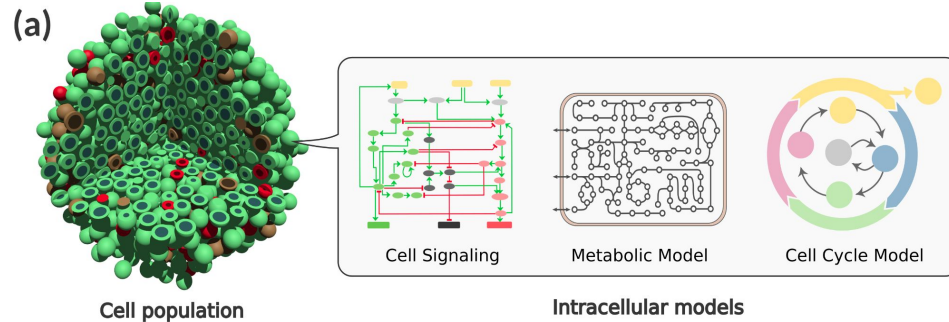
Microenvironment complexity

- Nutrient and oxygen gradients
- Cell-cell interactions
- Extracellular matrix
- Immune and stromal cells
- Spatial heterogeneity



Time complexity (adaptation/evolution)

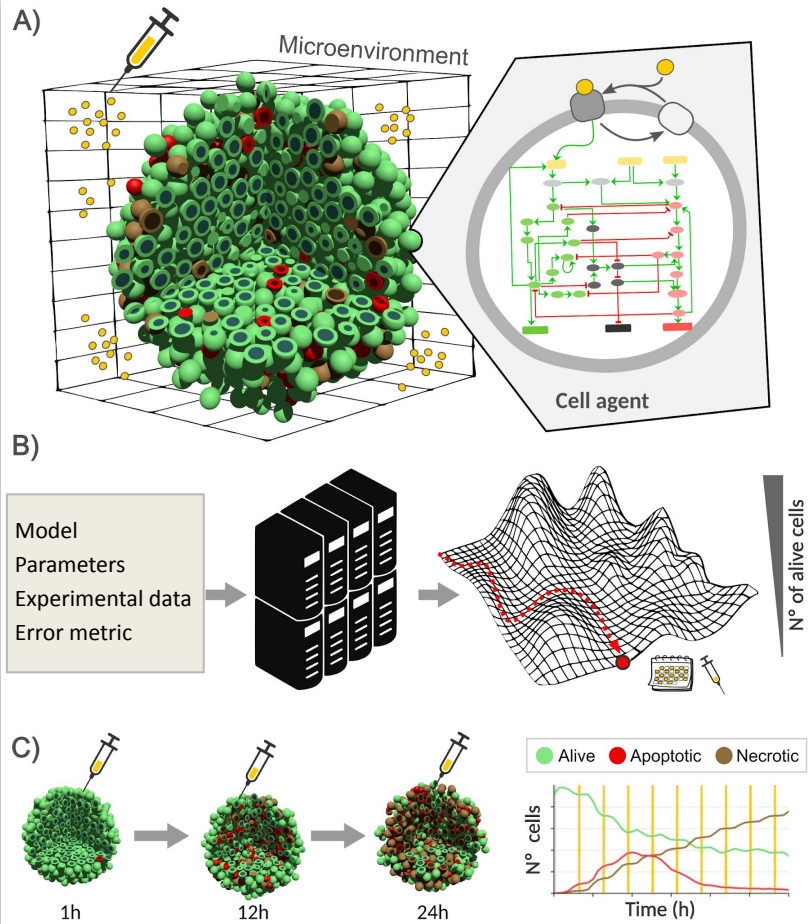
- Clonal evolution
- Treatment response
- Emergence of resistance
- Dynamic microenvironment
- Multiple time scales



Mechanistic multi-scale models can couple intracellular networks to cell-cell interactions, spatial microenvironment and temporal dynamics

Calibrate: making the model reproduce reality

Parameter inference through simulation-based optimization



Model Definition Define the biological system Identify critical parameters Formulate the model (equations, networks, rules)

Model Calibration Use experimental data as calibration targets Find parameter values that reproduce experimental observations Minimize simulation-data mismatch

Model Exploration Explore intervention strategies Optimize end-user goals e.g. reduce tumour size with minimum dosage

A different kind of hard

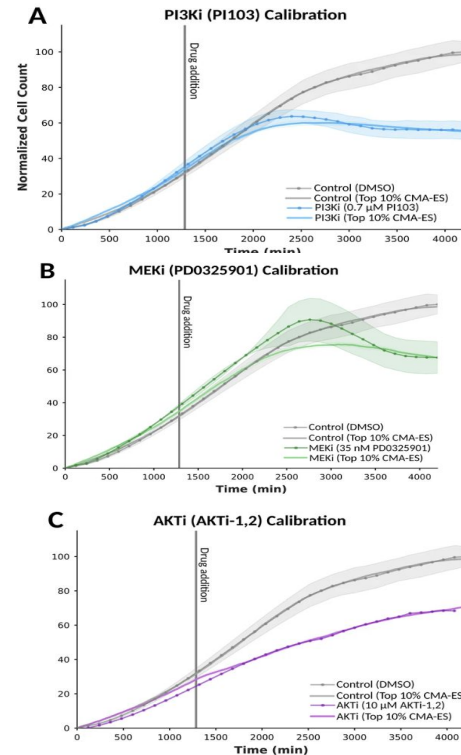
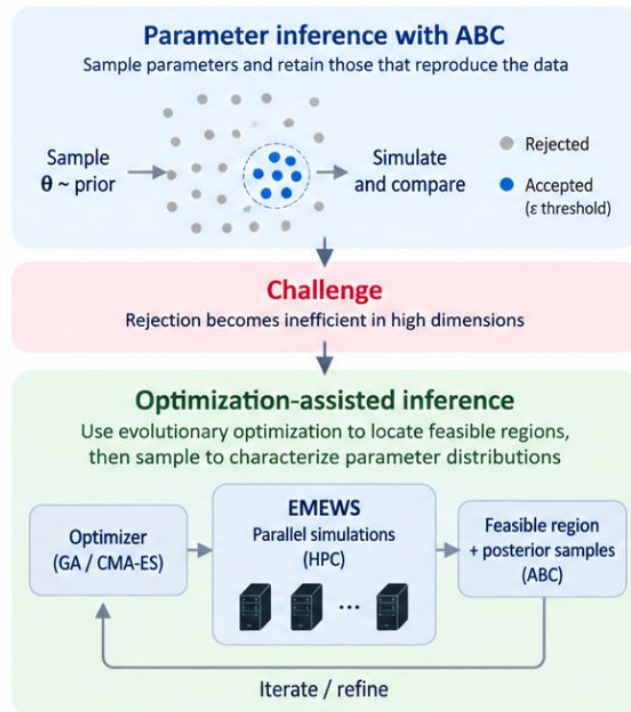
Simulation-based optimization: expensive evaluations and multiple good fits

High-dimensional space 18 Parameters

Expensive Simulations ABC + EMEWS
(HPC)

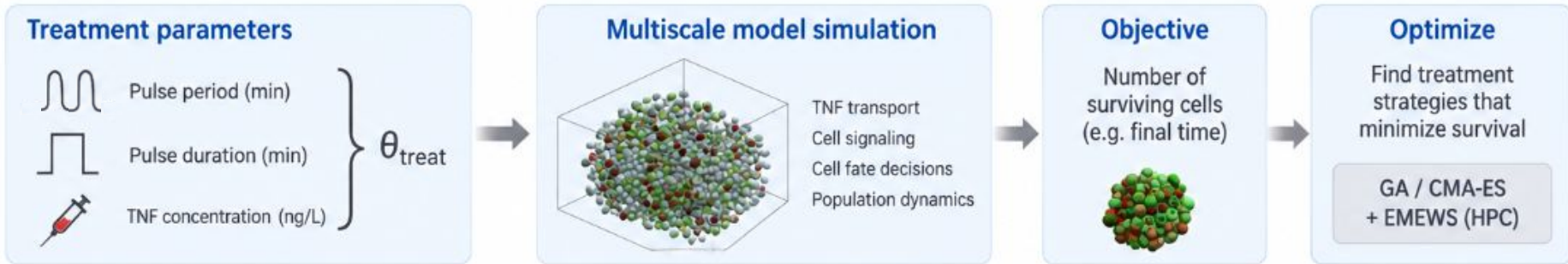
Parameter distribution Experimental
vs simulation

Parameter group	Examples
Growth	proliferation, apoptosis
Drug transport	diffusion, internalization
Drug action	binding, inhibition
Cell response	phenotype modulation
BN integration	signaling strength
Scaling	Boolean to continuous scaling

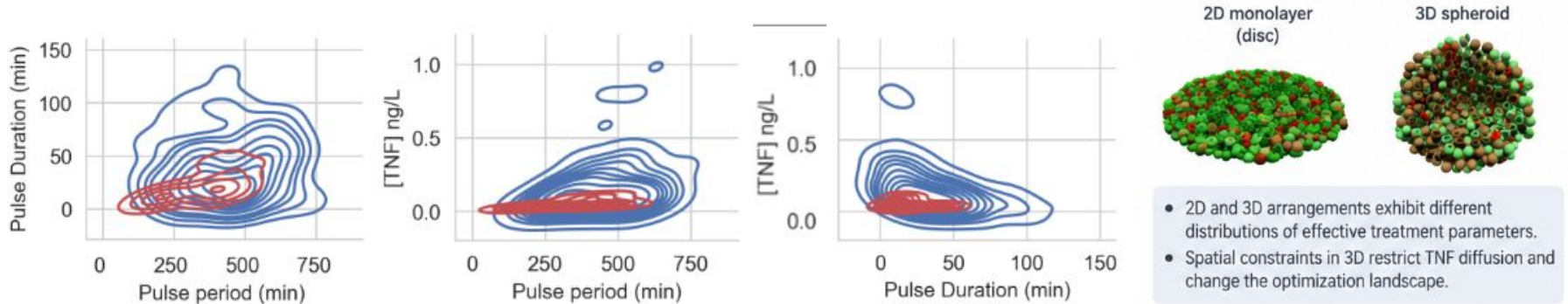


Optimizing Interventions

Which treatment strategy should we apply?



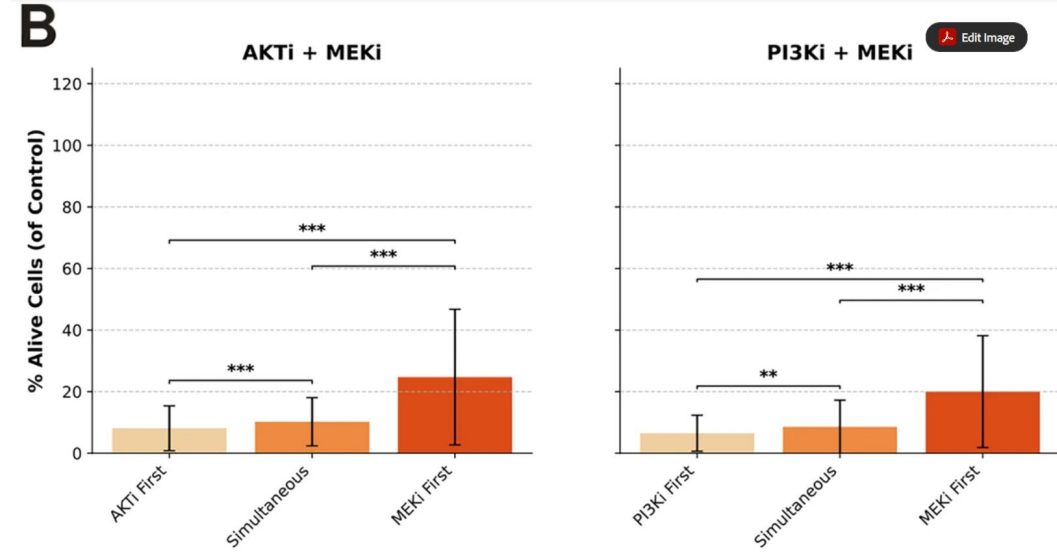
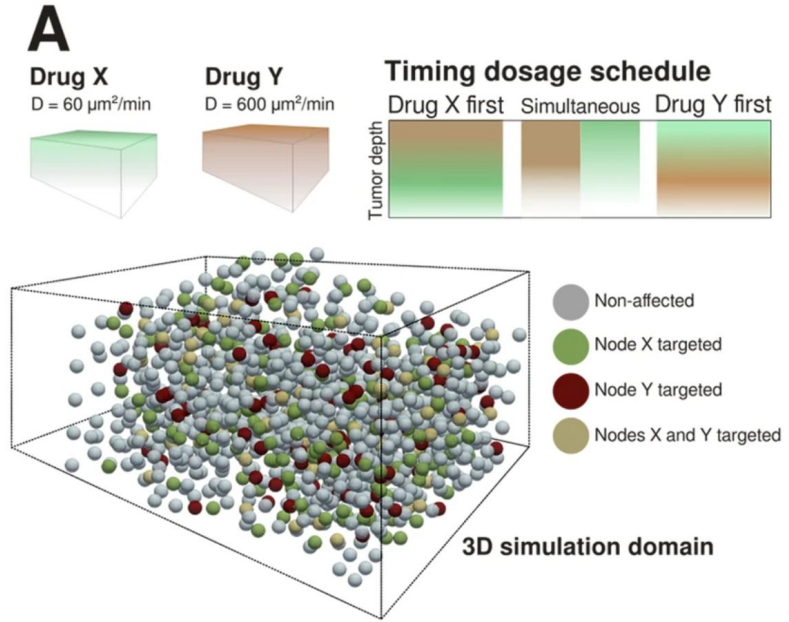
Effective treatment parameters differs for 2D and 3D cell arrangements



The optimization landscape is shaped by the biological model

The intervention space grows quickly

Discrete choices × continuous doses × temporal decisions



WHICH DRUGS? PI3Ki · MEKi · AKTi

HOW MUCH? dose · ratio

WHEN? simultaneous · sequential · duration

Different optimization problems at each step

Three optimization problems, one mechanistic workflow

From network structure to parameters to intervention

	INFER	CALIBRATE	INTERVENE
Question	What network?	What parameters?	What treatment?
Decision variables	Binary / discrete	Continuous / mixed	Discrete + continuous + temporal
Evaluation	Constraints + objective	Simulation $\leftrightarrow$ data	Simulation $\rightarrow$ response
Main challenge	Combinatorial structure	Expensive evaluations	Expensive + mixed search
Solutions	Multiple networks	Multiple good fits	Multiple strategies

From biological models to optimization benchmarks

Biological realism + reproducibility + controllable difficulty

1. INSTANCE What defines a problem instance? **Network size · parameter dimension · treatment space · simulation cost**



2. OBJECTIVE + CONSTRAINTS What counts as a valid solution? **Biological feasibility · model-data fit · treatment efficacy**

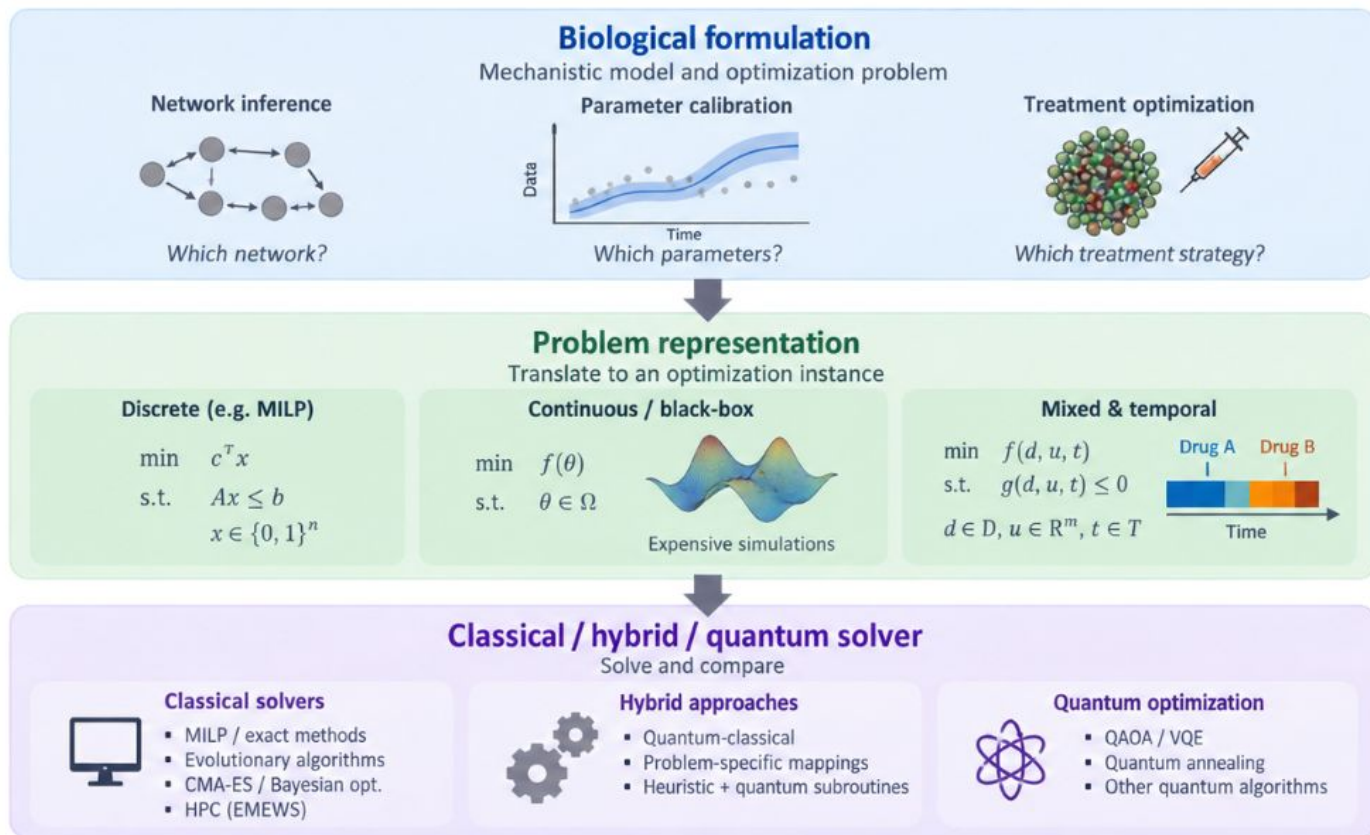


3. BASELINES What should we compare against? **MILP / exact methods · GA · CMA-ES · other classical heuristics**



4. METRICS How do we evaluate? **Objective · feasibility · runtime · evaluations · scaling**

Could these become quantum optimization benchmarks?



Key questions

- 1 Can we generate parameterized instances?**
Different sizes, dynamics, treatment spaces...
- 2 Can we control problem difficulty?**
Realistic and tunable complexity.
- 3 Can we define fair classical baselines?**
Reproducible instances and evaluation metrics.

Mechanistic biology can provide structured, realistic optimization problems.

Take-home messages

INFER → CALIBRATE → INTERVENE

- **What network?** *Discrete · combinatorial*
- **What parameters?** *Expensive · simulation-based*
- **What treatment?** *Mixed · temporal · simulation-based*

These are real biological problems — can we turn them into reproducible benchmarks for emerging optimization methods?

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Mechanistic modelling & motivation

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